

Importance Sampling



CHALMERS
UNIVERSITY OF TECHNOLOGY

Monte Carlo approximations

Two perspectives on Monte Carlo approximation

Given independent samples $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)} \sim p(\mathbf{x})$
we can approximate

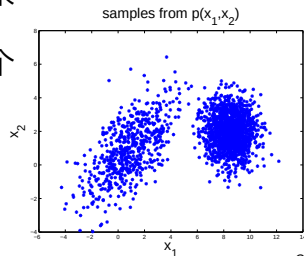
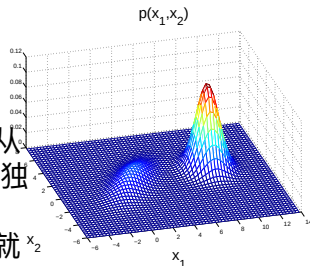
$$\mathbb{E}[\mathbf{g}(\mathbf{x})] \approx \frac{1}{N} \sum_{i=1}^N \mathbf{g}(\mathbf{x}^{(i)})$$

$$p(\mathbf{x}) \approx \frac{1}{N} \sum_{i=1}^N \delta(\mathbf{x} - \mathbf{x}^{(i)})$$

$$\mathbb{E}[\mathbf{g}(\mathbf{x})] = \int \mathbf{g}(\mathbf{x}) p(\mathbf{x}) d\mathbf{x} \approx \int \frac{1}{N} \sum_{i=1}^N \mathbf{g}(\mathbf{x}) \delta(\mathbf{x} - \mathbf{x}^{(i)}) d\mathbf{x} = \frac{1}{N} \sum_{i=1}^N \mathbf{g}(\mathbf{x}^{(i)})$$

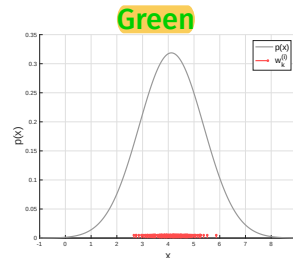
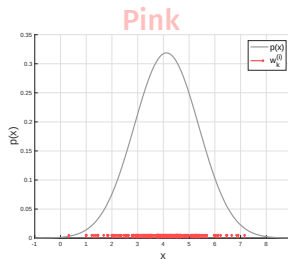
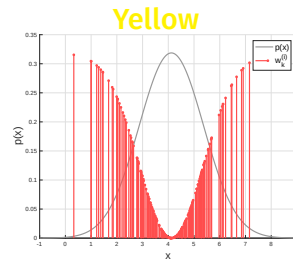
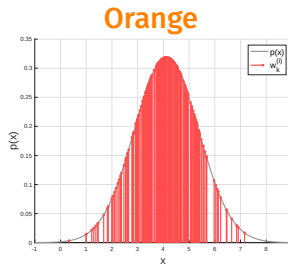
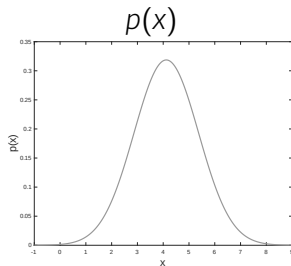
如果我们能直接从
目标分布 $p(\mathbf{x})$ 中独
立抽取大量样本

- (1) (粒子), 我们就
可以用这些样本来
近似计算期望值,
(2) 或者近似表示这个
分布本身



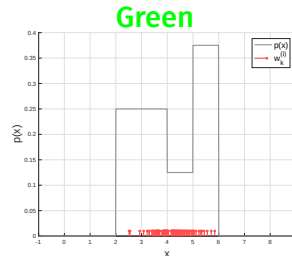
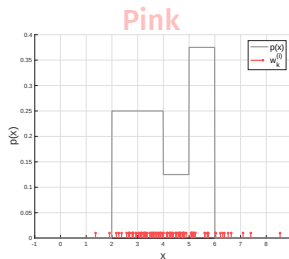
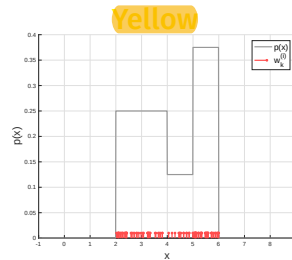
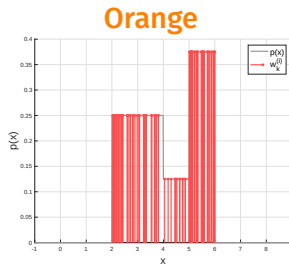
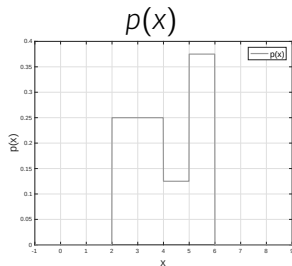
WHICH IS THE MONTE CARLO APPROXIMATION OF $p(x)$

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WHICH IS THE MONTE CARLO APPROXIMATION OF $p(x)$?

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Importance sampling

Importance sampling approximation to $p(\mathbf{x})$

- Generate samples, $\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \dots, \mathbf{x}^{(N)}$, from $q(\mathbf{x})$ and set

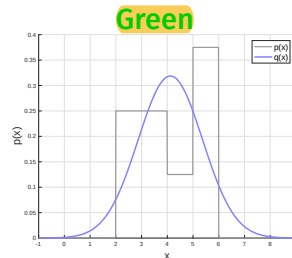
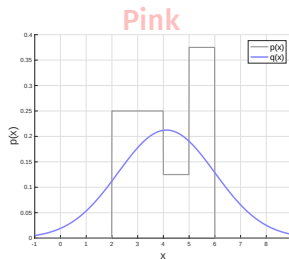
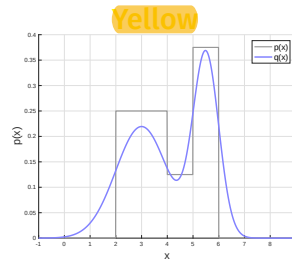
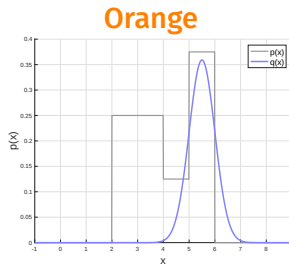
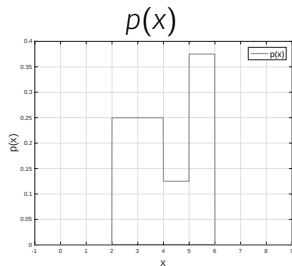
$$p(\mathbf{x}) \approx \sum_{i=1}^N w^{(i)} \delta(\mathbf{x} - \mathbf{x}^{(i)})$$

where

$$w^{(i)} = \frac{\tilde{w}^{(i)}}{\sum_{n=1}^N \tilde{w}^{(n)}} \quad \text{and} \quad \tilde{w}^{(i)} = \frac{p(\mathbf{x}^{(i)})}{q(\mathbf{x}^{(i)})}.$$

WHICH IS AN APPROPRIATE PROPOSAL DENSITY FOR $p(x)$?

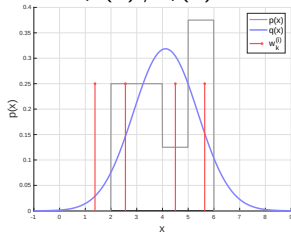
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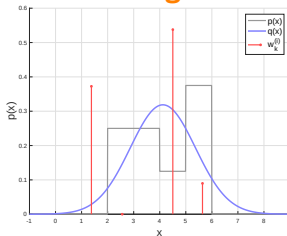
WHICH IS THE IMPORTANCE SAMPLING APPR. OF $p(x)$ USING $q(x)$?

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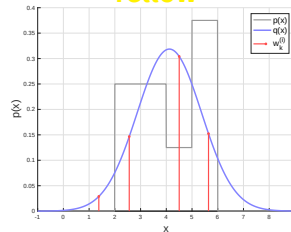
$p(x) / q(x)$



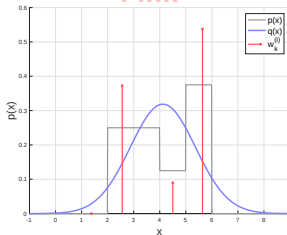
Orange



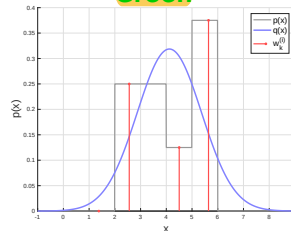
Yellow



Pink

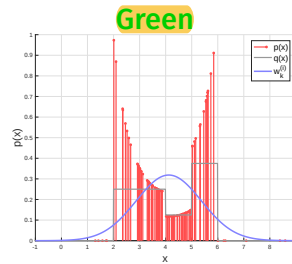
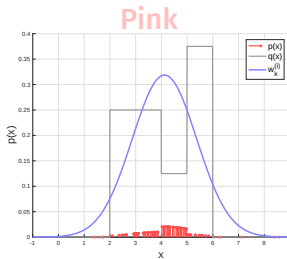
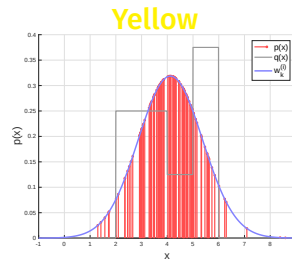
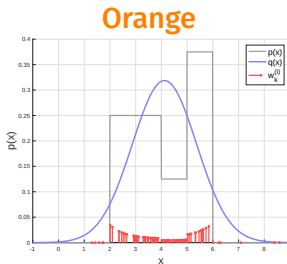
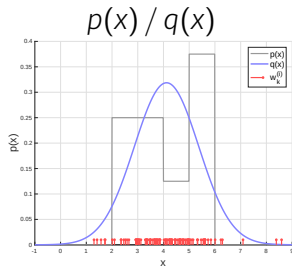


Green

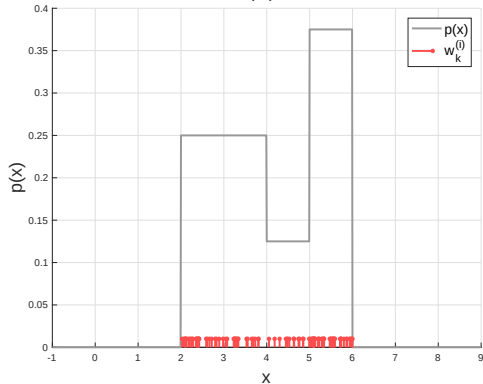


WHICH IS THE IMPORTANCE SAMPLING APPR. OF $p(x)$ USING $q(x)$?

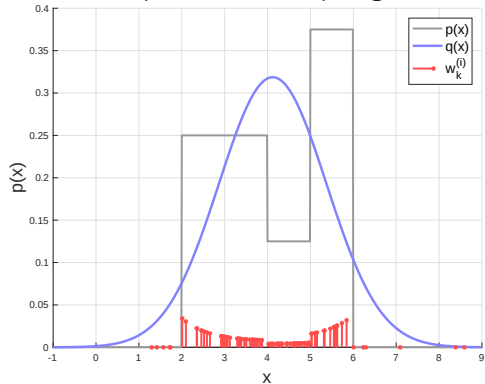
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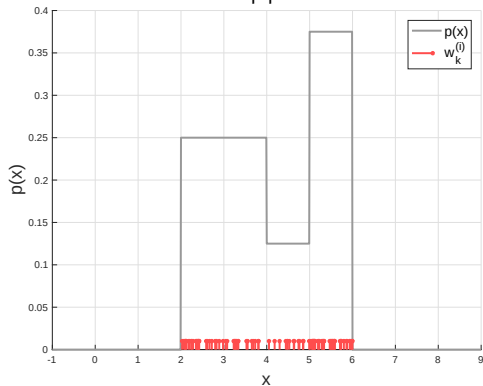
Monte Carlo approximation



Importance sampling



Monte Carlo approximation



Importance sampling /w resampling

